

References (Fourier + Wavelet)

Brunton, S. (2023). *Fourier analysis (data-driven science and engineering)* [Playlist]. Youtube.

https://youtube.com/playlist?list=PLMrJAKhleNNT_Xh3Oy0Y4LTj0Oxo8GqsC&si=b76Sxa80prYeDgfm

Cerna, M., & Harvey, A. F. (2000). *The fundamentals of FFT-based signal analysis and Measurement*. National Instruments.

chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.sjsu.edu/people/burford.furman/docs/me120/FFT_tutorial_NI.pdf

Everett, M. E. (2013). *Near-surface applied geophysics*. Cambridge University Press.

Kirsanov, A. (2022, August 15). *Wavelets: a mathematical microscope* [Video]. YouTube.

<https://youtu.be/jnxqHcObNK4?si=JN1AanzCTXucBsfe>

Mike X Cohen. (2019, December 20). *Welch's method for smooth spectral decomposition*

[Video]. YouTube. <https://youtu.be/YK1F0-3VvQI?si=z9uN4eL8oAOmHQiB>

Muradeli, J., Van Merriënboer, B., & Bondesson, D. (2025). *sssqueeze* (0.6.6) [Computer software]. GitHub. <https://github.com/OverLordGoldDragon/sssqueeze>

National Instruments. (2025, October 28). *LabVIEW advanced signal processing toolkit user manual: Short-time Fourier transform*.

<https://www.ni.com/docs/en-US/bundle/labview-advanced-signal-processing-toolkit/page/short-time-fourier-transform.html#:~:text=The%20best%20window%20length%20depends,to%20obtain%20a%20smoother%20STFT>

Newman, M. (2022). *Understand the output of the FFT* [Playlist]. Youtube.

https://www.youtube.com/playlist?list=PLWMUMyAoIbNv9c9qMoksBWv_YDn57EJHz

Polikar, R. (1996, June 5). *The wavelet tutorial part 1: Fundamental concepts & an overview of the wavelet theory*. Iowa State University.

<https://ccrma.stanford.edu/~unjung/mylec/WTpart1.html>

Polikar, R. (n.d.). *Wavelet tutorial - part 3*. Rowan University.

<https://users.rowan.edu/~polikar/WTpart3.html>

Siemens. (2029, August 29). *Wavelets / Time frequency analysis*.

<https://community.sw.siemens.com/s/article/wavelets-time-frequency-analysis>

Siemens. (2019, August 29). *Window correction factors*.

<https://community.sw.siemens.com/s/article/window-correction-factors>

References (Transfer Functions)

Åström, K. J., & Murray, R. M. (2004). *Feedback systems: An introduction for scientists and engineers* (pp. 139-166). California Institute of Technology.

chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.cds.caltech.edu/~murray/courses/cds101/fa04/caltech/am04_ch6-3nov04.pdf

GeeksforGeeks. (2025, July 23). *Cross-correlation analysis in Python*.

<https://www.geeksforgeeks.org/machine-learning/cross-correlation-analysis-in-python/>

Marinov, D. (2023, August 11). *Part 5: Polyphase FIR filters*. VHDLwhiz.

<https://vhdlwhiz.com/part-5-polyphase-fir-filters/>

Numiqo. (2023, March 21). Pearson correlation (simply explained) [Video]. YouTube.

<https://youtu.be/k7lctLRiZmo?si=nq4zMhB6c4sMm4ct>

RØDE. (2026). *Wireless PRO user guide | Getting started*.

<https://rode.com/en-ca/user-guides/wireless-pro/getting-started>

Vibration Research. (n.d.). *What is the cross spectral density (CSD)?*

<https://vibrationresearch.com/blog/what-is-cross-spectral-density-csd/>

VR University. (n.d.). *Coherence*.

<https://vru.vibrationresearch.com/lesson/coherence-signal-analysis/>

VR University. (n.d.). *Coherence Graph*.

<https://vru.vibrationresearch.com/lesson/coherence-graph/>

VR University. (n.d.). *Transfer Function*.

<https://vru.vibrationresearch.com/lesson/transfer-function/>

References (Grilla Project)

Konstantinovskiy, T. (2024, July 22). *Introduction to the Savitzky-Golay filter: A comprehensive guide (using python)*. Medium.

<https://medium.com/pythoneers/introduction-to-the-savitzky-golay-filter-a-comprehensive-guide-using-python-b2dd07a8e2ce>

United States Environmental Protection Agency. (2025, Dec 16). *Passive seismic (HVSr)*.

<https://www.epa.gov/environmental-geophysics/passive-seismic-hvsr>

Xu, R., & Wang, L. (2021). The horizontal-to-vertical spectral ratio and its applications.

EURASIP Journal on Advances in Signal Processing, 2021, Article 75.

<https://link.springer.com/article/10.1186/s13634-021-00765-z>

References (Laser Timer)

Arduino. (2024, Nov 13). *If your board is not detected by arduino IDE*.

<https://support.arduino.cc/hc/en-us/articles/4412955149586-If-your-board-is-not-detected-by-Arduino-IDE>

Ehnert, L. (n.d.). *Assembly instructions*.

<https://sites.google.com/larryrocktimer.com/larryrocktimer/home/assemblyinstructions?authuser=0>

Ehnert, L. (n.d.). *Electronic parts*.

<https://sites.google.com/larryrocktimer.com/larryrocktimer/home/electronic-parts?authuser=0>

Ehnert, L. (n.d.). *Larry rock timer*.

<https://sites.google.com/larryrocktimer.com/larryrocktimer/home?authuser=0>

Ehnert, L. (n.d.). *Operating instructions*.

<https://sites.google.com/larryrocktimer.com/larryrocktimer/home/operatinginstructions?authuser=0>

Ehnert, L. (n.d.). *Plastic parts*.

<https://sites.google.com/larryrocktimer.com/larryrocktimer/home/plasticparts?authuser=0>

Seeed Studio. (2024, Aug 14). *Getting started with seeed studio xiao esp32c3*.

https://wiki.seeedstudio.com/XIAO_ESP32C3_Getting_Started/

TechWithRita. (2023, Dec 3). *Getting started with the xiao esp32-c3 simple blink program*

[Video]. Youtube. https://youtu.be/JTkKJ7wTRvA?si=Cymid7_82cfj6Ca6

Documentation Sites

ArduinoDocs. (n.d.). *Language reference*. <https://docs.arduino.cc/language-reference/#functions>

Librosa. (n.d.). *Core IO and DSP*. <https://librosa.org/doc/0.11.0/core.html>

Librosa. (n.d.). *Display*. <https://librosa.org/doc/0.11.0/display.html>

Matplotlib. (n.d.). *API reference*. <https://matplotlib.org/stable/api/index>

Matplotlib. (n.d.). *Using matplotlib*. <https://matplotlib.org/stable/users/index.html>

Muradeli, J. (2025, August 2). *ssqueezepy* [Computer software]. GitHub.

<https://github.com/OverLordGoldDragon/ssqueezepy>

NumPy. (2023, September 16). *NumPy reference*.

<https://numpy.org/doc/1.26/reference/index.html#reference>

NumPy. (n.d.). *NumPy user guide*. <https://numpy.org/doc/1.26/user/index.html#user>

Pandas. (n.d.). *API reference*. <https://pandas.pydata.org/docs/reference/index.html>

Pandas. (n.d.). *User guide*. https://pandas.pydata.org/docs/user_guide/index.html

Plotly. (n.d.). *Python API reference for plotly*. <https://plotly.com/python-api-reference/>

PyWavelets. (n.d.). *API reference*. <https://pywavelets.readthedocs.io/en/latest/ref/index.html>

Scikit-Learn. (n.d.). *API reference*. <https://scikit-learn.org/stable/api/index.html>

SciPy. (n.d.). *SciPy API*. <https://docs.scipy.org/doc/scipy/reference/index.html#scipy-api>

SciPy. (n.d.). *SciPy user guide*. <https://docs.scipy.org/doc/scipy/tutorial/index.html#user-guide>

Other

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